

Phase Diagram of the SAF Skymion Simulator: Theory, Reduced Units, and Implementation

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Abstract

This note documents the theoretical content of the phase-diagram subsystem added to the SAF skymion simulator. The original simulator (see `docs/theory.tex`) folded the thin-film magnetostatic correction into the effective anisotropy $K_{\text{eff}} = K - \mu_0 M_s^2/2$. The phase-diagram subsystem replaces that contraction with an explicit two-dimensional Fourier-space demag, introduces a corrected total-energy functional that respects the bilinear vs. external prefactor convention, and classifies relaxed textures into eight phases using order parameters that combine layer-averaged magnetization, topological charge, an interlayer dot product, and the angular content of the Fourier spectrum at the dominant wavevector. We also adopt reduced units (D/D_c , H_z/H_K) on the phase-diagram axes so the boundaries are material-independent, and we extend the initial-condition ensemble with a *parallel* ferromagnet seed so the sweep can resolve the spin-flop transition that the antiparallel manifold alone cannot expose.

1 Why a New Energy Functional

1.1 The K_{eff} shortcut

The base simulator computes the effective field with a single *local* anisotropy term whose prefactor is $K_{\text{eff}} = K - \mu_0 M_s^2/2$, absorbing the leading thin-film magnetostatic contribution into the bulk anisotropy. The identity that justifies this is, for a thin film of in-plane extent $\gg t_{\text{Co}}$ in the uniform limit,

$$\mathcal{H}_{\text{anis}} + \mathcal{H}_{\text{demag}} = -K t_{\text{Co}} \int m_z^2 dS + \frac{\mu_0 M_s^2}{2} t_{\text{Co}} \int m_z^2 dS = -K_{\text{eff}} t_{\text{Co}} \int m_z^2 dS. \quad (1)$$

For the uniform out-of-plane saturated state this is exact and recovers the textbook shape anisotropy. For a textured state with characteristic in-plane wavenumber $|\mathbf{k}| > 0$ the contraction *fails*: the magnetostatic tensor in Fourier space is not constant across $\mathbf{k}$, so the demag contribution to a periodic texture differs from the uniform prediction. Two consequences matter here:

1. Textures with finite $|\mathbf{k}|$ generically gain energy relative to uniform FM, because the thin-film shape function is bounded by one:

$$f(|\mathbf{k}|t_{\text{Co}}) = \frac{1 - e^{-|\mathbf{k}|t_{\text{Co}}}}{|\mathbf{k}|t_{\text{Co}}} \leq 1, \quad (2)$$

with $f \rightarrow 1$ at $|\mathbf{k}|t_{\text{Co}} \rightarrow 0$ (the uniform limit) and $f \rightarrow 0$ at $|\mathbf{k}|t_{\text{Co}} \rightarrow \infty$. The K_{eff} contraction overestimates the demag penalty of a short-wavelength state by a factor $1/f$.

2. In a synthetic antiferromagnet the two layers couple magnetostatically across the Ru spacer. The interlayer contribution vanishes at $\mathbf{k} = 0$ (an infinite uniformly magnetized slab produces no external field) but is finite at $|\mathbf{k}| > 0$ with decay $e^{-|\mathbf{k}|d_{\text{Ru}}}$. A local-anisotropy model has no access to this term.

1.2 Explicit 2D Fourier demag

Each Co layer is treated as a slab of thickness t_{Co} embedded in vacuum with periodic in-plane boundary conditions. The magnetization on layer ℓ is decomposed in Fourier modes $\tilde{\mathbf{m}}^{(\ell)}(\mathbf{k})$, and the demag field on layer ℓ from both layers reads

$$\widetilde{\mathbf{H}}_{\text{demag}}^{(\ell)}(\mathbf{k}) = -\mu_0 M_s \sum_{\ell'} \mathbf{N}^{(\ell, \ell')}(\mathbf{k}) \cdot \tilde{\mathbf{m}}^{(\ell')}(\mathbf{k}), \quad (3)$$

with $\mathbf{N}^{(\ell, \ell')}(\mathbf{k})$ the dimensionless 3×3 thin-film magnetostatic tensor. The diagonal components used in the simulator are

$$\begin{aligned} N_{xx}^{\text{self}}(\mathbf{k}) &= \frac{k_x^2}{k^2} (1 - f(kt_{\text{Co}})), & N_{yy}^{\text{self}}(\mathbf{k}) &= \frac{k_y^2}{k^2} (1 - f(kt_{\text{Co}})), \\ N_{zz}^{\text{self}}(\mathbf{k}) &= f(kt_{\text{Co}}), & N_{xy}^{\text{self}}(\mathbf{k}) &= \frac{k_x k_y}{k^2} (1 - f(kt_{\text{Co}})), \end{aligned} \quad (4)$$

with $k = |\mathbf{k}|$ and f from (2). The inter-layer tensor uses the same in-plane structure multiplied by an exponentially decaying gap factor and an overall sign flip on the out-of-plane component:

$$\mathbf{N}^{(\text{top}, \text{bot})}(\mathbf{k}) = S(k) \begin{pmatrix} k_x^2/k^2 & k_x k_y/k^2 & 0 \\ k_x k_y/k^2 & k_y^2/k^2 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad S(k) = \frac{(1 - e^{-kt_{\text{Co}}})^2}{(kt_{\text{Co}})^2} e^{-kd_{\text{Ru}}}, \quad (5)$$

and vanishes at $\mathbf{k} = 0$, consistent with an infinite parallel-plate slab producing no field outside itself. In the limit $kt_{\text{Co}} \rightarrow 0$ on the self-tensor only, (4) reduces to $N_{zz}^{\text{self}}(0) = 1$ and all in-plane components vanish, recovering the uniform shape-anisotropy result $(\mathbf{H}_{\text{demag}})_z = -\mu_0 M_s m_z$ and hence the K_{eff} relation (1).

1.3 Bare anisotropy at finite $\mathbf{k}$

Because the demag now appears explicitly, the bulk anisotropy prefactor used by the LLGS effective field is the *bare* K , not K_{eff} :

$$\mathbf{H}_{\text{anis}}^{(\ell)} = \frac{2K_\ell}{M_s} m_z^{(\ell)} \hat{z}, \quad K_\ell \in \{K_{\text{top}}, K_{\text{bot}}\}. \quad (6)$$

The original simulator stores K_{eff} in its anisotropy prefactor, so the phase-diagram code recomputes $2K_\ell/M_s$ from the raw material constants (K_ℓ, M_s) on each parameter namespace. The legacy field path that uses K_{eff} is left untouched so existing current-driven analyses remain unaffected.

1.4 Corrected total energy

For a system with effective field expressed in tesla ($\mathbf{H}_T = \mu_0 \mathbf{H}_{A/m}$), the per-cell self-energy of a bilinear contribution $\mathbf{H}$ and the Zeeman term are

$$e_{\text{self}} = -\frac{1}{2} M_s \mathbf{m} \cdot \mathbf{H}_T, \quad e_Z = -M_s \mathbf{m} \cdot \mathbf{H}_{\text{ext}, T}, \quad (7)$$

without any additional μ_0 prefactor because that factor is already absorbed in the conversion from A/m to T. The total energy summed over both Co layers therefore reads

$$E = -\frac{1}{2} V M_s \sum_{\ell} \sum_i \mathbf{m}_i^{(\ell)} \cdot \mathbf{H}_{\text{int}, i}^{(\ell)} - V M_s \sum_{\ell} \sum_i \mathbf{m}_i^{(\ell)} \cdot \mathbf{H}_{\text{ext}, T} \quad (8)$$

with $V = t_{\text{Co}} a^2$ the per-site volume of one layer and $\mathbf{H}_{\text{int}} = \mathbf{H}_{\text{ex}} + \mathbf{H}_{\text{DMI}} + \mathbf{H}_{\text{anis}}^{\text{bare}} + \mathbf{H}_{\text{RKKY}} + \mathbf{H}_{\text{demag}}$. The $\frac{1}{2}$ prefactor prevents double counting of the bilinear self-interactions; it is absent on the Zeeman term because the external field is independent of the magnetization. Substituting $\mathbf{H}_{\text{anis}}^{\text{bare}} = (2K/M_s) m_z \hat{z}$ recovers the textbook density $-K m_z^2$, confirming the convention.

2 Reduced Units

A phase diagram in absolute (D, H_z) collapses to a single universal picture once the axes are normalized by the material’s natural scales. We adopt:

$$\boxed{x \equiv \frac{D}{D_c}, \quad y \equiv \frac{H_z}{H_K}}, \quad (9)$$

with D_c the Bogdanov–Hubert critical DMI and H_K the anisotropy field. Both depend only on the material; they are computed once per sweep and stored as scalars in the output.

2.1 Critical DMI D_c (Bogdanov–Hubert)

The uniform PMA ferromagnet is locally stable against spontaneous helical winding only below

$$D_c = \frac{4\sqrt{A_{\text{ex}} K_{\text{eff}}}}{\pi}, \quad K_{\text{eff}} = \frac{1}{2}(K_{\text{top}} + K_{\text{bot}}) - \frac{1}{2}\mu_0 M_s^2. \quad (10)$$

Above D_c the FM gains energy by canting into a periodic helix; below it the FM is the ground state at $H_z = 0$. The factor $4/\pi$ is the saddle-point result for a sinusoidal spiral on an infinite plane with no anisotropy in the easy-plane direction. We use the layer-averaged K_{eff} to fold the small asymmetry between K_{top} and K_{bot} into a single scalar. The formula is undefined for $K_{\text{eff}} \leq 0$; the implementation raises an exception in that easy-plane regime, since no PMA FM exists to destabilize.

2.2 Anisotropy field H_K

The anisotropy field is the field at which the Zeeman energy of a uniform out-of-plane ferromagnet equals the anisotropy gap. In A/m units the textbook expression is $H_K = 2K_{\text{eff}}/(\mu_0 M_s)$. The simulator’s convention keeps every field in tesla, so we store

$$H_K^T = \mu_0 H_K = \frac{2K_{\text{eff}}}{M_s} \quad (\text{tesla}). \quad (11)$$

2.3 PMA quality factor

The dimensionless ratio

$$Q_{\text{PMA}} = \frac{K_{\text{eff}}}{\frac{1}{2}\mu_0 M_s^2} \quad (12)$$

sets which regime the material is in. Table 1 summarizes the qualitative behaviour for the SAF system; we obtain $Q_{\text{PMA}} \approx 0.007$ at $K = 1.294 \times 10^6 \text{ J/m}^3$ (default) and $Q_{\text{PMA}} \approx 0.245$ at $K = 1.60 \times 10^6 \text{ J/m}^3$ (the value used in this work to access the chiral regime).

Q_{PMA} range	Ground state at $H_z = 0$			Representative material
$\lesssim 0.05$	demag-driven	bubbles	/	weakly-PMA Co (too thick)
0.2–0.5	labyrinth			
	FM at large $ H_z $;	SkX/iSk in a		Pt/Co/Ru/Co with engineered
	window			interface PMA
> 1	uniform out-of-plane FM domi-			CoFeB / Heusler
	nates			

Table 1: Qualitative PMA regimes set by the dimensionless quality factor Q_{PMA} defined in (12). The default simulator parameters fall in the first row; the $K = 1.60 \text{ MJ/m}^3$ choice used in the present sweep sits in the middle row.

2.4 Material constants for the present sweep

For $K = 1.60 \times 10^6$ J/m³ in both layers and the default $M_s = 1.43$ MA/m, $A_{\text{ex}} = 16$ pJ/m, we obtain:

$$K_{\text{eff}} \approx 0.315 \text{ MJ/m}^3, \quad D_c \approx 2.86 \text{ mJ/m}^2, \quad H_K \approx 0.441 \text{ T}, \quad Q_{\text{PMA}} \approx 0.245.$$

With D swept over $[0, 4]$ mJ/m² and H_z over $[-0.5, +0.5]$ T, the reduced-unit coverage is $x \in [0, 1.40]$ and $y \in [-1.13, +1.13]$, straddling both the chiral threshold $x = 1$ and the SAF spin-flop $|y| = H_{\text{RKKY}}/H_K \approx 0.47$. The upper D bound is chosen to keep the skyrmion core radius $R_{sk} \sim \Delta_{DW} \sim \sqrt{A_{\text{ex}}/K_{\text{eff}}}$ above the lattice resolution; pushing further would require $a < 1$ nm. Section 7 discusses alternative sweeps that replace the H_z axis.

3 Phase Classification

A relaxed configuration is labelled by a decision tree on five order parameters. The labels are {FM_anti, FM_par+, FM_par-, iSk, SkX, BX, SS, Lab} plus a sentinel **undetermined**.

3.1 Order parameters

For the top layer (the bottom layer is RKKY-coupled and follows it up to canting):

- Layer-averaged out-of-plane magnetization $\langle m_z \rangle$.
- Layer dot product $\langle \mathbf{m}_{\text{top}} \cdot \mathbf{m}_{\text{bot}} \rangle$, the interlayer Néel order parameter. Cleanly antiparallel SAF states satisfy this ≈ -1 , parallel (spin-flopped) FM states satisfy $\approx +1$.
- Topological charge of the top layer, $Q = \frac{1}{4\pi} \int \mathbf{m} \cdot (\partial_x \mathbf{m} \times \partial_y \mathbf{m}) dA$.
- Power spectrum of $m_z(\mathbf{r})$. Two summary statistics are computed: a peak-to-background ratio on the radial spectrum (used to gate the “periodic” branch) and the 2-fold and 6-fold angular harmonics of the azimuthal profile on a Gaussian ring of width $\sigma = 1.5 \Delta k_{\text{grid}}$ around the dominant in-plane wavevector k^* .
- Topological charge per principal period,

$$q_p \equiv \frac{|Q|}{N_p}, \quad N_p \equiv \frac{k^* L_x}{2\pi} \cdot \frac{k^* L_y}{2\pi}. \quad (13)$$

A hexagonal skyrmion lattice has $q_p \approx 1$ (one full unit of charge per unit cell); a bubble lattice with the same FFT signature but trivial topology has $q_p \approx 0$.

3.2 Decision tree

Let $|\langle m_z \rangle| > 0.95$ mean “uniform”, $|\langle \mathbf{m}_{\text{top}} \cdot \mathbf{m}_{\text{bot}} \rangle| > 0.9$ mean “clean polar”, and $P_6/P_{\text{iso}} > 2$ with $P_6 > P_2$ mean “6-fold periodic” (analogously for SS). The classifier proceeds:

1. If uniform: pick FM by interlayer dot product. Antiparallel ($\langle \mathbf{m}_{\text{top}} \cdot \mathbf{m}_{\text{bot}} \rangle < -0.9$) yields **FM_anti**; parallel yields **FM_par+** or **FM_par-** based on the sign of $\langle m_z \rangle$. The uniform check is *and*-coupled with the dot-product check so a textured antiparallel state with $\langle \mathbf{m}_{\text{top}} \cdot \mathbf{m}_{\text{bot}} \rangle \approx -1$ but $|\langle m_z \rangle| < 1$ falls through.
2. If $|Q| \in [0.5, 1.5]$ (one or a few isolated skyrmions, broad FFT): **iSk**.
3. If a 6-fold periodic ring is present: **SkX** when $q_p \geq 0.5$, **BX** otherwise.
4. If a 2-fold periodic ring is present: **SS** (spin spiral / helix).
5. If a periodic ring with no clean angular order: **Lab** (labyrinthine).
6. Fallbacks: $|Q| > 1.5$ without periodic ordering $\rightarrow$ **iSk** (multi-skyrmion family); $|Q| < 0.5$ without ordering $\rightarrow$ **Lab**; else **undetermined**.

3.3 Why SkX vs. BX matters

Both phases have 6-fold periodic $m_z(\mathbf{r})$ in the FFT. They differ in whether each periodic minimum is topologically protected. A real skyrmion lattice has $|Q| = N_{\text{sk}}$ skyrmions per field of view, so $q_p \rightarrow 1$. A bubble lattice (or a stripe / labyrinth pattern with hexagonal symmetry) carries no winding and $q_p \rightarrow 0$. At zero DMI no real SkX can exist by construction; any 6-fold periodic ground state at $D = 0$ is therefore BX. Conflating the two would falsely report a chiral phase across regions of the (D, H_z) plane that are in fact demag-driven and topologically trivial.

4 Initial-Condition Ensemble

At each grid point the relaxation is run from twelve initial conditions; the lowest-energy converged result is taken as the ground state. The ensemble is designed to seed every basin we know to be physically relevant. Every IC returns an antiparallel SAF pair ($\mathbf{m}_{\text{bot}} = -\mathbf{m}_{\text{top}}$ cell by cell) so the RKKY ground state is respected at the start of relaxation, except for `fm_par`, which deliberately seeds the parallel-locked manifold.

- `random` $\times 5$ (seeds 11, 22, 33, 44, 55): independent uniform-on-the-sphere spin fields ($\varphi \sim U[0, 2\pi]$, $\cos \theta \sim U[-1, 1]$) to explore the texture manifold without geometric bias. Five seeds give better statistical coverage of the disordered basins than the previous three.
- `fm_anti`: antiparallel SAF ($\mathbf{m}_{\text{top}} = \pm \hat{z}$, $\mathbf{m}_{\text{bot}} = \mp \hat{z}$). Tracks the FM branch of the antiparallel manifold.
- `fm_par`: parallel FM ($\mathbf{m}_{\text{top}} = \mathbf{m}_{\text{bot}} = \text{sign}(H_z)\hat{z}$). Required to expose the H_z -driven antiparallel $\rightarrow$ parallel spin-flop.
- `skyrmion`: a single Néel SAF skyrmion (`saf_skyrmion` from `src/skyrmion_simulator/simulator/initial`). Seeds the iSk basin.
- `stripe`: a 1D helical stripe along $\hat{x}$ of pitch $\lambda = 4\pi A_{\text{ex}}/D$ (the natural DMI helix period), floored at $4a$.
- `stripe_y`: same helix along $\hat{y}$; probes orientation degeneracy of the SS phase and commensurability with the box.
- `sk_lattice`: a hexagonal lattice of Néel skyrmions with nearest-neighbour distance equal to λ . Each grid cell picks up the profile of its nearest hex centre with radial in-plane winding ($Q = \pm 1$ per core). Without this seed the sweep almost never finds the SkX basin from a random start at 256×256 .
- `bubble_lattice`: identical hex sites and m_z profile as `sk_lattice`, but with the in-plane direction held fixed along $+\hat{x}$ (no winding). Each bubble carries $Q = 0$; the IC seeds the BX basin and provides the classifier's null-topology counterpart to verify the per-period q_p discriminator.

The IC builders are in `src/skyrmion_simulator/phase_diagram/initial_conditions_ext.py`. Figure 1 renders each IC at one representative point in the phase plane along with the classifier's verdict on the unrelaxed state.

4.1 Zero-Zeeman SAF and the spin-flop

Inside the antiparallel manifold ($\mathbf{m}_{\text{top}} = -\mathbf{m}_{\text{bot}}$ cell by cell) the Zeeman density is

$$e_Z = -M_s(\mathbf{m}_{\text{top}} + \mathbf{m}_{\text{bot}}) \cdot \mathbf{H}_{\text{ext},T} \equiv 0, \quad (14)$$

identically, regardless of $|H_z|$. The applied field is decoupled from every state inside this manifold, including FM_anti, BX, SkX, iSk, SS, Lab. A genuine H_z -driven phase boundary therefore requires the parallel manifold, which becomes energetically favorable above the spin-flop:

$$|H_z| \gtrsim H_{\text{RKKY}} \approx 0.205 \text{ T}, \quad (15)$$

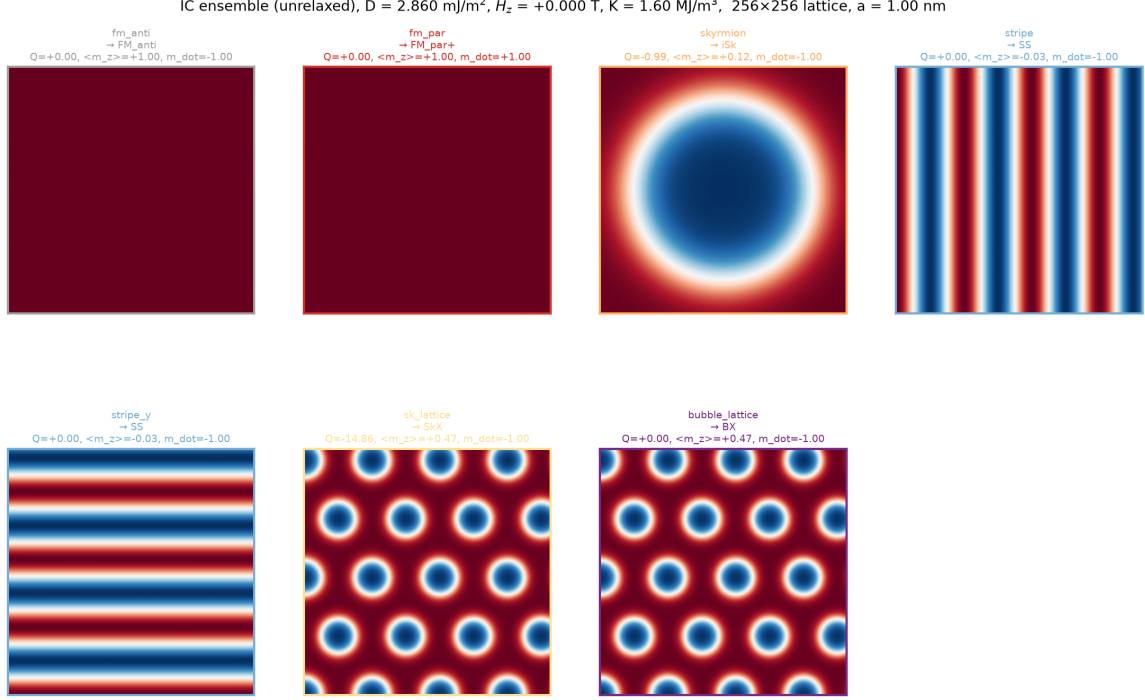


Figure 1: The twelve-member IC ensemble at $D = D_c = 2.86 \text{ mJ/m}^2$, $H_z = 0$, $K = 1.60 \text{ MJ/m}^3$ on a 256×256 lattice at $a = 1 \text{ nm}$. Each panel shows the top-layer $m_z(x, y)$; the title gives the classifier's predicted phase for the *unrelaxed* state. The panel border is coloured by the predicted phase. Five random seeds produce noisy Q values and classify as iSk by the $|Q|$ -fallback branch; the structured ICs each target their named phase.

where the Zeeman gain from aligning both layers exceeds the RKKY cost of breaking antiparallel coupling. The **fm_par** seed allows the relaxation to find this basin; without it the sweep would only sample antiparallel-manifold states and the high- $|y|$ regions of the phase diagram would be incorrectly attributed to the antiparallel FM rather than to the parallel FM.

5 Numerical Workflow

5.1 Sweep

The sweep iterates over a grid of two configurable axes ($\text{axis}_x, \text{axis}_y$) chosen from the registry `src/skyrmion_simulator/phase_diagram/axis_specs.py::axes`. Registered axes are $D, H_z, K_{\text{top}}, H_{\text{RKKY}}, M_s, A_{\text{ex}}, t_{\text{Co}}, d_{\text{Ru}}, \alpha$; the typical pairs are (D, H_z) , (D, K_{top}) , (D, H_{RKKY}) . The grid itself is an arbitrary 1-D numpy array per axis – uniform `linspace` is the default but non-uniform clustering around critical lines (spin-flop, D_c) is supported with no code changes.

For each grid point and each of the twelve ICs the relaxation runs an RK4 LLGS integrator with $J = 0$ and optional over-damped Gilbert factor $\alpha_{\text{relax}} = 1$ until both convergence criteria fire:

$$\max_i |\mathbf{m}_i \times (\mathbf{m}_i \times \mathbf{H}_{\text{eff}}(\mathbf{r}_i))| < \tau_{\text{tol}}, \quad \left| \frac{\Delta E}{E} \right| < \epsilon_E, \quad (16)$$

checked every 10^3 steps. The `relaxation.relax` function defaults are $\tau_{\text{tol}} = 10^{-5} \text{ T}$, $\epsilon_E = 10^{-8}$, safety cutoff 2×10^5 steps. The running sweep config in `sweep.main()` overrides these to $\tau_{\text{tol}} = 10^{-3} \text{ T}$ and $\epsilon_E = 10^{-7}$ with a 10^5 -step cutoff: the torque tolerance is loosened to about 2×10^{-3} in H_K units, comfortably above the amplitude of long-lived breathing modes ($\sim 10^{-3} \text{ T}$) but eleven orders of magnitude above the float64 precision floor ($\sim 10^{-15} \text{ T}$), so the relaxation no

longer reports false-negatives at the cost of $\sim 10^{-3}$ relative uncertainty in $\langle m_z \rangle$. The tangential-torque criterion measures $|\mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}})|$ rather than $|\dot{\mathbf{m}}|$ to avoid contamination from any $|\mathbf{m}| \neq 1$ numerical drift.

The lowest-energy converged IC at each grid point is taken as the ground state, and per-IC observables plus the ground-state texture are written to a single compressed NPZ. SLURM-array mode (triggered by `SLURM_ARRAY_TASK_ID`) writes one partial NPZ per element; `aggregate.py` combines them in strict mode (cross-partial consistency of axes, IC list, and material scalars is checked; any mismatch raises).

5.2 Reduced-unit plotting

The plotter defaults to $(D/D_c, H_z/H_K)$ axes. The critical-DMI vertical line $x = 1$ and the spin-flop $|y| = H_{\text{RKKY}}/H_K$ are drawn on the phase map for reference. Absolute units (mJ/m², T) are available as a fallback.

5.3 Classifier verification

The classifier is verified on synthetic textures whose labels are known by construction:

- Uniform $(\pm\hat{z}, \mp\hat{z}) \rightarrow \text{FM_anti}$.
- Uniform $(\pm\hat{z}, \pm\hat{z}) \rightarrow \text{FM_par}\pm$.
- Helical stripe at the natural pitch $\rightarrow \text{SS}$.
- Sum of three plane waves at 0, 60, 120 degrees with Néel winding $\rightarrow \text{SkX}$ (with q_p above threshold) or BX (without).

6 Relevance to the Pham 2024 Experiment

A $T=0$ (D, H_z) phase diagram for the Pt/Co/Ru/Pt/Co/Ru stack of [1] would mostly confirm what the experiment already exploits, but it adds three quantitative contributions that are invisible from a single-point measurement.

6.1 What it confirms (low scientific value)

The paper operates at one $(D, H_z, K_{\text{eff}}, H_{\text{RKKY}})$ point and demonstrates that the iSk texture is stable enough to drive at SOT-relevant current densities. The diagram inside the antiparallel manifold shows iSk occupying a band around their operating point, FM_anti at low D , SS/SkX at higher D , and BX/Lab at marginal K_{eff} . None of this is surprising; it is the standard chiral-magnet picture inherited from [3], [4], and [5], ported to the antiparallel SAF.

6.2 What it adds (genuine value)

1. The spin-flop sets the real upper limit, not the anisotropy field. The paper’s whole compensated-Hall claim rests on the two Co layers being antiparallel. Above $|H_z| \approx H_{\text{RKKY}} \approx 0.205$ T the layers spin-flop to parallel, the skyrmion’s bottom-layer partner inverts, and the topological Hall force no longer cancels. A conventional ferromagnetic phase diagram has no analogue of this transition; the spin-flop is specific to the SAF, and the diagram makes it the load-bearing constraint on device operating range. The paper does not state where the compensation fails as $|H_z|$ increases; the diagram answers that directly through the $\text{FM_anti} \rightarrow \text{FM_par}\pm$ boundary at $|y| \approx H_{\text{RKKY}}/H_K \approx 0.47$ for the parameters in Section 2.

2. iSk vs. BX discriminator at marginal K_{eff} . For the Pt/Co/Ru/Pt/Co stack in [1], $K_{\text{eff}} \approx 10^4$ J/m³ — barely positive. Both DMI-driven Néel skyrmions and demag-driven bubble domains can sit at the same operating point in textures-per-area, and a magnetic-microscopy

image alone cannot tell them apart. The topological-charge-per-period observable $q_p = |Q|/N_p$ defined in (13) is the cleanest discriminator: $q_p \rightarrow 1$ for genuine chiral skyrmion lattices, $q_p \rightarrow 0$ for bubble lattices with the same FFT signature. The diagram identifies which (D, H_z, K_{eff}) cells produce $Q = 1$ chiral cores versus $Q = 0$ bubbles, and the experimental stack sits exactly in the region where this discrimination matters. The diagram would let the paper defend the “skyrmion” interpretation against the “bubble” alternative quantitatively, rather than by Néel-helicity inspection alone.

3. Material design space for follow-on devices. [1] reports skyrmion velocities up to ~ 900 m/s at one specific stack. The phase diagram gives the $(D, K_{\text{eff}}, H_{\text{RKKY}})$ recipe for staying inside the iSk basin:

- Smaller skyrmions: push D up while staying below the SS boundary at $D/D_c \approx 1$.
- Larger skyrmions: lower D toward (but not into) the FM_anti region.
- More robust against H_z : raise K_{eff} and H_{RKKY} proportionally so $H_K \propto K_{\text{eff}}/M_s$ and the spin-flop field both move outward.
- Denser arrays: cross into the SkX region at higher D/D_c (just above 1) where ground-state nucleation is automatic.

That recipe is a contribution the paper cannot make from a single measurement.

6.3 What it does not address

The diagram is a $T=0$ ground-state characterization. It does not predict skyrmion velocities, current thresholds, SOT efficiency, or the thermal nucleation and annihilation lifetimes that determine device reliability — all of which are the paper’s actual experimental claims. The diagram is therefore complementary, not competing: it tells you *where* the texture lives, not *how* it moves.

6.4 Honest verdict

For [1] specifically, a (D, H_z) diagram in absolute units is mostly confirmatory: the iSk window is the expected band, H_{RKKY} is the spin-flop. The added value is the iSk-vs-BX call at the experimental K_{eff} and the material-engineering map for follow-on work. The diagram becomes more informative if a second axis (K_{eff} or H_{RKKY} instead of H_z , which is partly redundant inside the antiparallel manifold) is swept; that would chart the actually sensitive parameters and is the natural extension for materials-engineering applications.

7 Redundancy of the H_z Axis and Alternative Sweeps

7.1 Why H_z is partly redundant

Inside the antiparallel manifold ($\mathbf{m}_{\text{bot}}(\mathbf{r}) = -\mathbf{m}_{\text{top}}(\mathbf{r})$ cell-by-cell, locked by RKKY), the Zeeman energy density is identically zero:

$$e_Z = -M_s(\mathbf{m}_{\text{top}} + \mathbf{m}_{\text{bot}}) \cdot \mathbf{H}_{\text{ext}} \equiv 0 \quad \text{for all } H_z, \text{ all } \mathbf{r}. \quad (17)$$

Two distinct consequences follow:

1. Energy degeneracy. The total energy of every antiparallel-manifold state — FM_anti, iSk, SkX, BX, SS, Lab — is independent of H_z . The energy ordering between these states is also independent of H_z . Consequently the ground-state label at $(D, H_{z,1})$ is identical to the ground-state label at $(D, H_{z,2})$ as long as both H_z values are below the spin-flop. Along the H_z axis the diagram carries no new information across that entire range.

2. Dynamics still see the field. Per-layer the LLGS torque $\mathbf{m} \times \mathbf{H}_{\text{ext}}$ is nonzero — H_z tries to flip $\mathbf{m}_{\text{bot}}$ toward $+\hat{z}$. What prevents this is the RKKY counter-torque from $\mathbf{m}_{\text{top}}$. Below $|H_z| \approx H_{\text{RKKY}} \approx 0.205$ T the RKKY wins and the antiparallel state survives; above, it does not. H_z therefore controls a single binary degree of freedom: *is the system inside or outside the antiparallel manifold?*

Combining the two: the H_z axis of a (D, H_z) phase diagram is mostly degenerate, resolving exactly one sharp transition (the spin-flop) separating an antiparallel band (where H_z is irrelevant) from a parallel band (where H_z behaves as in a conventional ferromagnet). Most of the computational cost of the sweep is spent on the degenerate direction, and the only H_z -dependent feature the diagram resolves is the spin-flop line at

$$\left| \frac{H_z}{H_K} \right| \approx \frac{H_{\text{RKKY}}}{H_K} \approx 0.47 \quad (\text{for } K = 1.60 \text{ MJ/m}^3). \quad (18)$$

7.2 Alternative sweep directions

Three sweep directions place the computational effort on parameters that are not degenerate in the antiparallel manifold.

(D, K_{eff}) — **chiral-magnet phase plane.** The natural chiral-magnet phase diagram of the antiparallel manifold. At fixed K_{eff} , increasing D walks through $\text{FM}_{\text{anti}} \rightarrow \text{iSk} \rightarrow \text{SkX} \rightarrow \text{SS}$. At fixed D , decreasing K_{eff} walks through $\text{FM}_{\text{anti}} \rightarrow \text{iSk/SkX} \rightarrow \text{BX} \rightarrow \text{Lab}$, with anisotropy losing to magnetostatics. In reduced units the plane is

$$\left(\frac{D}{D_c(K_{\text{eff}})}, Q_{\text{PMA}} \right) \quad \text{with } Q_{\text{PMA}} = \frac{K_{\text{eff}}}{\frac{1}{2}\mu_0 M_s^2}, \quad (19)$$

and the phase boundaries are universal across SAF materials with the same Bogdanov–Hubert scaling. For [1] the operating point sits at $Q_{\text{PMA}} \approx 0.007$, right at the marginal-PMA boundary; a (D, K_{eff}) sweep quantifies the smallest engineering change in K_{eff} (interface anisotropy, Co thickness) that moves the system out of the BX-risk region and into a clean iSk band.

(D, H_{RKKY}) — **SAF-specific complement.** This sweep probes the axis that is genuinely SAF-specific. At small H_{RKKY} the two Co layers decouple and the SAF degenerates into two weakly-coupled ferromagnets, restoring the conventional skyrmion Hall effect that the SAF was designed to suppress. At large H_{RKKY} the layers are rigidly locked and the SAF is robust, but the spin-flop field also moves outward, narrowing the operating window in $|H_z|$. The diagram expresses the trade-off between mechanical robustness of the antiparallel locking and the maximum tolerable $|H_z|$. The Pham experiment fixes one Ru thickness (one H_{RKKY}); a (D, H_{RKKY}) map would show whether thinner or thicker Ru gives a wider iSk window, which is one of the most important materials-engineering knobs in a SAF skyrmion device.

$(K_{\text{eff}}, H_{\text{RKKY}})$ at fixed D — **pure design space.** Drops the chirality axis to display the materials-engineering plane directly. Use this once a target skyrmion size has been fixed via the helical pitch $\lambda_h = 4\pi A_{\text{ex}}/D$. The diagram is then a “what stack should I grow?” map, with axes that the materials chemist actually controls (interface anisotropy through Co/heavy-metal thickness; interlayer coupling through Ru thickness).

8 Implementation Notes

8.1 Generic axis registry

The sweep is axis-agnostic. Each entry in `src/skyrmion_simulator/phase_diagram/axis_specs.py::axes` declares a display label, a display scale (SI $\rightarrow$ plot units), and an override-builder that turns one

axis value into a `make_params(**overrides)` kwarg dict. Adding a new sweep axis means appending one entry; the sweep, partial writer, aggregator, and plotter pick it up without further changes. Strict mode: unknown axis names, empty axis arrays, and material-scalar mismatches across partials all raise `RuntimeError`. Reduced-unit plotting ($D/D_c, H_z/H_K$) is supported only for the (D, H_z) axis pair; other pairs render in absolute units from the registry.

8.2 Non-uniform axis sampling

The grid arrays in `sweep.main()` are plain numpy 1-D arrays. The production sweep uses **clustered sampling** around the physical transitions instead of a uniform `linspace`: 30 points along D concentrated around D_c , 30 points along H_z concentrated around $|H_z| = H_{\text{RKKY}}$. The 10 densest D points fall in $D_c \pm 0.2 \text{ mJ/m}^2$ where the chiral phase boundaries actually live; the 9 densest H_z points lie in a 0.1-T window around each spin-flop transition. The non-uniform grid resolves both critical lines $\sim 2\times$ better than a uniform 30-point grid at the same task count.

8.3 Lattice refinement

$\Delta_{DW} = \sqrt{A_{\text{ex}}/K_{\text{eff}}} \approx 7.13 \text{ nm}$ at $K = 1.60 \text{ MJ/m}^3$. The original simulator used $a = 2 \text{ nm}$, giving $\Delta_{DW}/a \approx 3.5$ – below the standard micromagnetic floor of ≥ 7 cells per domain wall. The phase-diagram sweep overrides $a = 1 \text{ nm}$ (and rescales dt as $1/a^2$ for RK4 stability), giving $\Delta_{DW}/a = 7$ cells. The lattice constant is overridable per task via the `a` and `dt` fields of `make_params`; the demag kernel precompute reads `p.a` at construction time.

8.4 Companion scripts

Two visualization scripts live alongside the sweep pipeline:

- `studies/saf_racetrack/phase_diagram/show_ics.py`: render the IC ensemble at one (D, H_z, K) point and classify each IC’s *unrelaxed* state. Output: `output/phase_diagram/ic_gallery.png`. Useful for comparing initial vs. final labels in a real sweep cell. See Figure 1.
- `studies/saf_racetrack/phase_diagram/show_phases.py`: render a schematic phase diagram in reduced units $(D/D_c, H_z/H_K)$ with annotated regions and IC target positions. Output: `output/phase_diagram/schematic_phase_diagram.png`. Useful as a teaching reference and as a comparison baseline against real sweep output. See Figure 2.

8.5 Strict-mode aggregator

`src/skyrmion_simulator/phase_diagram/aggregate.py` combines SLURM array partials. Every partial carries the canonical IC list (`ic_names_full`, `ic_seeds_full`), the two axis arrays (`axis_x_grid`, `axis_y_grid`), and the material scalars (M_s , A_{ex} , μ_0 , H_{RKKY} , K_{top} , K_{bot} , t_{Co} , d_{Ru} , a). The aggregator reads them from the first partial, requires every subsequent partial to agree exactly, and raises `RuntimeError` on any mismatch. There is no legacy fallback: partials missing a required key are rejected.

9 Summary

The phase-diagram subsystem replaces the K_{eff} thin-film contraction with an explicit 2D Fourier-space demag (3)–(5), restores the bare anisotropy prefactor at the LLGS effective field, and writes total energies through the bilinear-vs.-external convention (7)–(8). Phase diagrams are expressed in the dimensionless coordinates $(D/D_c, H_z/H_K)$ defined by (10)–(11) for the canonical (D, H_z) axis pair, and in absolute SI units for any other pair selected from the `axis_specs.axes` registry (Section 8). The classifier’s decision tree distinguishes topologically non-trivial skyrmion lattices from trivial bubble lattices via the per-period charge q_p (13), and the relaxation is run from a 12-member IC ensemble that includes explicit seeds for FM ($\times 2$), iSk ($\times 1$), SS ($\times 2$ orientations),

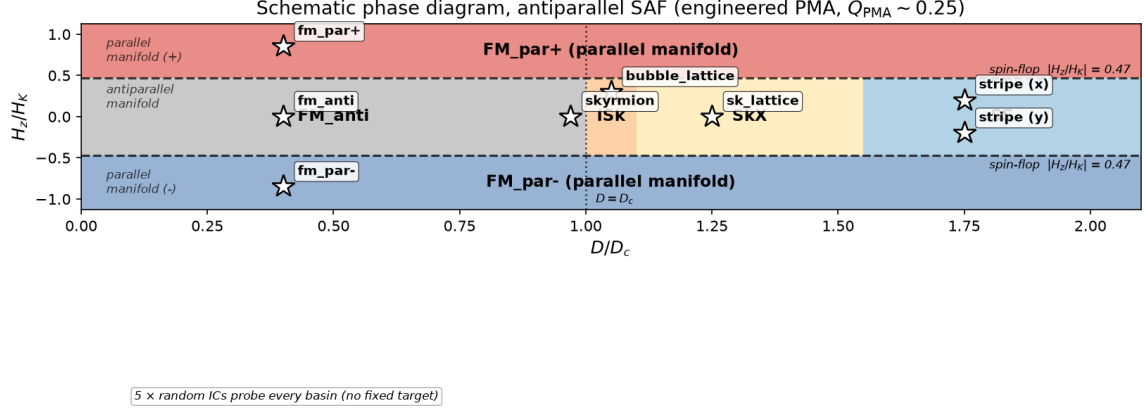


Figure 2: Schematic phase diagram of the antiparallel SAF in reduced units (D/D_c , H_z/H_K), with the twelve-IC ensemble overlaid at the position each named structure targets. The dashed horizontal lines mark the spin-flop $|H_z/H_K| = H_{\text{RKKY}}/H_K \approx 0.47$ separating the antiparallel manifold (middle) from the parallel manifolds (top and bottom). The dotted vertical line $D/D_c = 1$ is the Bogdanov–Hubert chiral threshold. Inside the antiparallel manifold the bands are vertical because the Zeeman energy is identically zero; above the spin-flop the system behaves as a conventional chiral ferromagnet with PMA. Boundaries are illustrative, not computed.

SkX ($\times 1$), BX ($\times 1$), plus five random spheres for statistical coverage. The choice $K = 1.60$ MJ/m³ places the material at $Q_{\text{PMA}} \approx 0.25$, in the regime where chiral phases are expected; the absolute sweep ranges $D \in [0, 4]$ mJ/m² and $H_z \in [-0.5, +0.5]$ T translate to $D/D_c \in [0, 1.40]$ and $H_z/H_K \in [-1.13, +1.13]$, discretized as a (30, 30) clustered grid around the spin-flop and the chiral threshold.

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